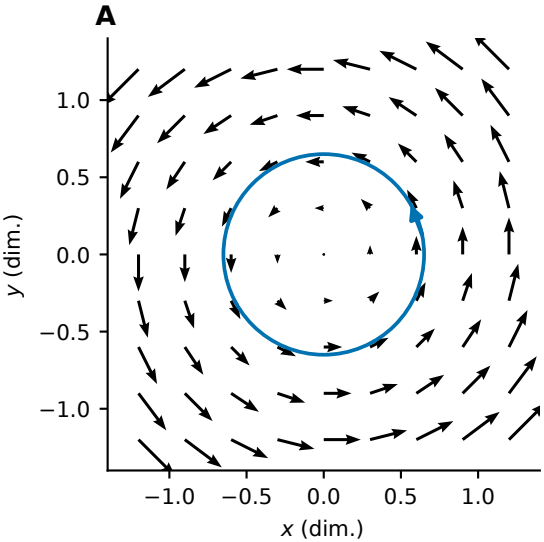
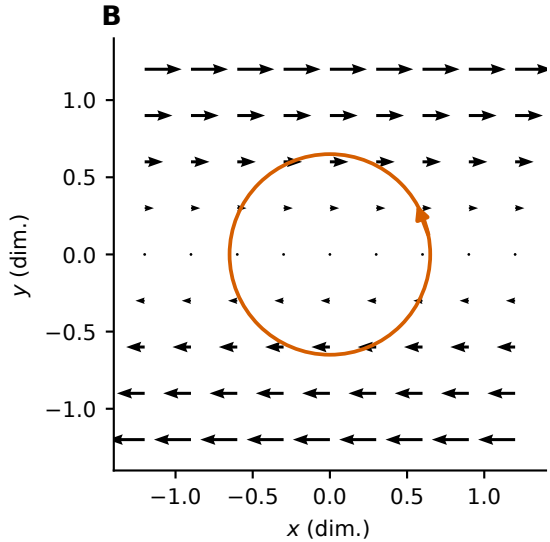
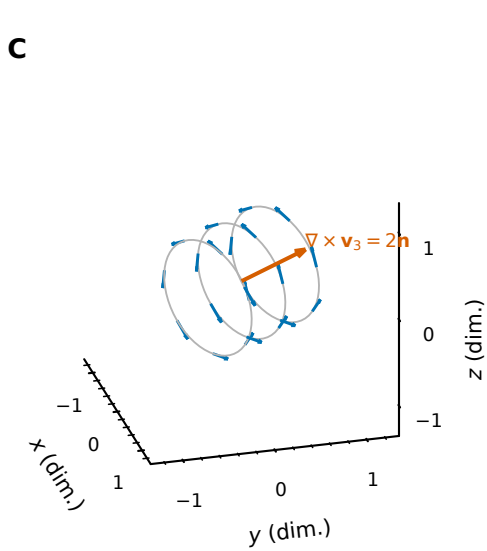


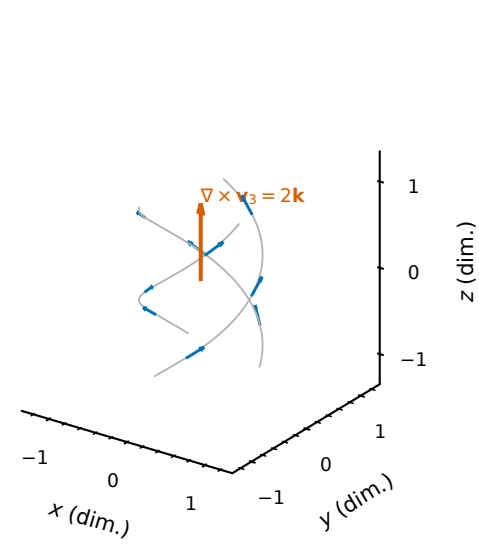
$\mathbf{v}_2 = (-y, x)^T$: $(\nabla \times \mathbf{v}_2)_z = 2$



$\mathbf{v}_2 = (y, 0)^T$: $(\nabla \times \mathbf{v}_2)_z = -1$



$\mathbf{v}_3 = \mathbf{n} \times \mathbf{r}$, $\mathbf{n} = (1, 1, 1)^T/\sqrt{3}$



$\mathbf{v}_3 = (-y, x, 1/2)^T$: rotation + translation